

HARSHA VARDHAN DOPPALAPUDI

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Education

International Institute of Information Technology Bangalore

July 2024 – May 2028

Bachelor of Technology in Data Science and Artificial Intelligence

3.37/4

Experience

NLP Research Intern (SRIP)

June 2026 – Present

IIIT Bangalore – Natural Language Processing Lab

Bangalore, Karnataka

- Co-developing a cross-cultural evaluation benchmark under Prof. Tulika Saha to quantify and mitigate Western bias in Large Language Models across **29 Indian regional cultures**.
- Engineered an automated data extraction pipeline in Python with the OpenAI API to parse dense anthropological texts, generating a structured dataset of **4,785 culturally grounded Rules-of-Thumb**.
- Designed an LLM-as-a-Judge validation framework that detects and regenerates hallucinated or duplicate entries, ensuring a high-fidelity, hallucination-free dataset for supervised fine-tuning (SFT).

Projects

CLV Intelligence Engine | Python, LightGBM, Optuna, Streamlit | [GitHub](#) | [Dashboard](#)

April 2025 – June 2025

- Engineered a stacked machine learning ensemble combining LightGBM with statistical BG/NBD probabilistic models, cutting **Mean Absolute Error by 43.6%** on **775K+ transactions**.
- Built a causal-inference T-Learner uplift model identifying **424 persuadable customers**, enabling ROI-maximizing allocation of an **INR 50L** quarterly retention budget.
- Served forecasts via a production Streamlit MLOps dashboard featuring **85.6%-coverage** conformal prediction intervals, automated PSI drift monitoring, and SHAP-based feature explainability.

Question Answering System | Python, PyTorch, Hugging Face | [GitHub](#) | [Live Demo](#)

March 2026 – April 2026

- Pre-trained a 12-layer, 768-dim custom Transformer encoder (110M params) from scratch via GPU-accelerated MLM on Wikipedia; fine-tuned on **130K+ SQuAD v2 QA pairs**.
- Achieved **58.2 F1** (Extractive) and **39.9 F1** (Generative) on SQuAD v2, outperforming from-scratch DistilBERT and T5-small baselines trained under identical conditions.
- Deployed an end-to-end inference pipeline as a containerized REST API on Hugging Face Spaces (Docker, FastAPI), serving FP16-quantized models optimized for CPU-based free-tier inference.

Traffic Violation Detection | YOLOv11, ONNX Runtime, FastAPI | [GitHub](#) | [Live Demo](#)

April 2026 – June 2026

- Built an end-to-end **two-wheeler violation detection** system — detecting no-helmet riders, overloaded bikes, and reading license plates — using a 3-stage YOLOv11 cascade (motorcycle → helmet → plate).
- Trained 3 specialized models on 17,670 curated images (COCO + VisDrone + Roboflow) on a remote RTX 4060 Ti GPU, achieving **mAP50 of 0.936** (plate), **0.829** (helmet), and **0.761** (motorcycle).
- Accelerated CPU inference **1.8×** via ONNX Runtime; deployed full-stack (FastAPI, Docker, GitHub Actions CI/CD) publicly on **Hugging Face Spaces**.

Technical Skills

Languages: Python, C++, SQL, Bash

ML Frameworks: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Optuna

ML/AI Areas: Machine Learning, Deep Learning, NLP, LLMs, Computer Vision, Reinforcement Learning

Methods: Fine-Tuning, Transfer Learning, Statistical Modeling, Causal Inference, A/B Testing, Conformal Prediction

Libraries & Tools: Hugging Face (Transformers, Datasets, Tokenizers), NumPy, Pandas, OpenCV, SHAP, Matplotlib, Jupyter

MLOps & Deployment: FastAPI, REST APIs, Docker, Streamlit, Git, Linux

Achievements

- Kaggle – Top 5% (Rank 48 / 1,344)** in Playground Series S6E6: Stellar Classification, using a multi-model ensemble of LightGBM, XGBoost, and CatBoost with balanced-accuracy optimization and protected anchor blending.
- Open-Source Contributor – EleutherAI/lm-evaluation-harness** (12.9K+ stars): contributed an IndicQA evaluation benchmark covering 11 Indian languages with custom F1/EM metric hooks ([PR #3829](#)); fixed a regex extraction bug in the Hendrycks Math LaTeX parser ([PR #3827](#)).